



Growth and climate change Stochastic IAMs in Dynare

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Summer School on Macroeconomic Modelling and Policy

Introduction

In what follows:

- ▶ I Introduce the basic conclusions that we can take from [Nordhaus1992](#)) from deterministic models:
 - ▶ Example of a deterministic/stochastic baby DICE
- ▶ Then I provide a short run perspective from [Sahuc et al.2025](#)) in which stabilization policies are also affected by mitigation policy in the short term
 - ▶ Example of a New Keynesian model

Why macroeconomists should care

- ▶ Climate change reshapes macroeconomic dynamics: productivity, inflation, employment.
- ▶ Monetary and fiscal policies are impacted through:
 - ▶ *Climateflation*: Damages reduce productivity, raise prices.
 - ▶ *Greenflation*: Carbon taxes raise costs during the transition.
- ▶ Macroeconomic tools must adapt: from stable-steady-state thinking to structural transitions.
- ▶ Need for integrated models that capture both physical and nominal frictions.

What is an Integrated Assessment Model (IAM)?

- ▶ IAMs link **economic activity** and **climate dynamics** to assess the costs and benefits of climate policies.
- ▶ The **IPCC** relies on a large pool of IAMs to produce long-term climate scenarios.
- ▶ These models are **highly heterogeneous**: many do not include optimization, forward-looking behavior, microfoundations, or expectations.
- ▶ The traditional benchmark is **DICE** (Nordhaus, 1992):
 - ▶ Embeds climate science into a **Ramsey-type growth model**.
 - ▶ Delivers optimal carbon pricing paths based on welfare maximization.

Why Climate Economics Needs a Short-Term View

- ▶ Policy institutions - central banks, treasuries, regulators - increasingly face the need to assess:
 - ▶ **Climate shocks** (e.g. disasters, weather volatility),
 - ▶ **Transition dynamics** (e.g. carbon tax, green investment),
 - ▶ **Stability risks** (price, financial, macroeconomic).
- ▶ This calls for models that are not only long-term and normative, but also **short-term and dynamic**.
- ▶ Yet, our standard macro tools are not designed for this:
 - ▶ DSGE models often focus on stationary dynamics near a steady state,
 - ▶ IAMs neglect shocks, expectations, and nominal frictions.
- ▶ This creates a **methodological gap** when addressing climate change as both a long-term trend and a short-term source of instability.

Outline

- 1 Introduction
- 2 A Long Term View: The DICE Model
- 3 A short term view: the New Keynesian Climate Model

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- ▶ DICE (Dynamic Integrated Climate-Economy model) was developed by [Nordhaus1992](#) and went through several (minor) updates up to the latest 2023 snapshot;
- ▶ Simple but powerful model to provide fast estimates of the social cost of carbon and transition pathways for the world economy;
- ▶ It has become an intuitive benchmark.
- ▶ Back in the 90s, Bill Nordhaus's work on climate has never been published in top 5 journals (always rejected) but got published in Sciences, PNAS, etc.

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 - ▶ Solow growth model with climate externality;
 - ▶ Centralized equilibrium based on Ramsey allocations;
 - ▶ Purely deterministic setup;
- ▶ Physical capital, temperatures and carbon boxes are the main state variables that the planner uses to optimally maximize welfare.
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Carbon cycle

- Atmospheric loading of CO_2 is a system of differential equations:

$$M_t = \Phi_M M_{t-1} + \Xi_M E_t, \quad (1)$$

where $M_t = [M_t^{AT}, M_t^{UP}, M_t^{LO}]'$ comprises 3 layers of carbon: atmospheric, upper and lower ocean respectively.

- Matrices are determined by:

$$\Phi_M = \begin{bmatrix} 1 - \phi_{12} & \phi_{21} & 0 \\ \phi_{12} & 1 - \phi_{21} - \phi_{23} & \phi_{32} \\ 0 & \phi_{23} & 1 - \phi_{32} \end{bmatrix} \text{ and } \Xi_M = \begin{bmatrix} \xi_M \\ 0 \\ 0 \end{bmatrix}$$

- Carbon emissions go into the atmosphere, before being captured by oceans through a long lasting process.

Temperatures

- Temperature anomalies $T_t = [T_t^{AT}, T_t^{OC}]$ of atmosphere and ocean respectively:

$$T_t = \Phi_T T_{t-1} + \Xi_T F_t, \quad (2)$$

- Matrices are given by:

$$\Phi_T = \begin{bmatrix} \varphi_{11} & \varphi_{12} \\ \varphi_{21} & \varphi_{22} \end{bmatrix} \text{ and } \Xi_T = \begin{bmatrix} \xi_T \\ 0 \end{bmatrix}$$

- Cooling effects of oceans through φ_{12} , ξ_T is the heating sensitivity to radiative forcing F_t .

Radiative forcing

- ▶ Greenhouse effect via radiative forcing (W/m^2):

$$F_t = \eta \log \left(M_t^{AT} / M_{1750} \right) / \log(2) + F_t^{EX} \quad (3)$$

- ▶ Here, η denote the forcing (Wm^{-2}) of equilibrium CO_2 doubling ($M_t^{AT} = 2M_{1750}$).
- ▶ F_t^{EX} is an exogenous process tracking the contribution of other sources of GHG (i.e. methane) on F_t

Input structure

- World population (billion):

$$L_t = L_{t-1} (L_T / L_{t-1})^{\ell_g}, \quad (4)$$

$\ell_g \in [0, 1]$ convergence rate, $L_T \in [0, +\infty)$ long term population.

- Law of motion of capital:

$$K_t = (1 - \delta_K) K_{t-1} + I_t, \quad (5)$$

$\delta_K \in [0, 1]$ is the depreciation rate.

Production of goods and emissions

- Production is Cobb-Douglas:

$$Y_t = A_t K_{t-1}^{\gamma} L_t^{1-\gamma} \quad (6)$$

where $\gamma \in [0, 1]$ is capital intensity, exogenous TFP A_t , physical capital K_{t-1} .

- Emissions (GtCO2):

$$E_t = \sigma_t (1 - \mu_t) Y_t + E_t^{\text{land}} \quad (7)$$

where σ_t decoupling rate, μ_t is abatement share, E_t^{land} is exogenous process of carbon emission from change in land use (i.e. deforestation).

Equilibrium in goods and climate

- Resource constraint:

$$Y_t \left(1 - \theta_{1,t} \mu_t^{\theta_2}\right) \Omega(T_t) = C_t + I_t \quad (8)$$

LHS is output loss from abatement and damage, RHS is demand side.

- Damages are given by:

$$\Omega(T_t) = 1/(1 + aT_t^2) \quad (9)$$

This expression is subject to deep uncertainty, and should be taken as a normative measure of the social loss for the society of rising temperatures. Tipping points could be introduced [[Weitzman2012](#)].

Exogenous trends

Total factor productivity:

$$\Delta \log A_t = g_A - \delta_a \log (A_t/A_0)$$

Decoupling rate of carbon emission to GDP:

$$\Delta \log \sigma_t = g_\sigma - \delta_\sigma \log (\sigma_t/\sigma_0)$$

Cost of abating carbon emissions:

$$\theta_{1,t} = (1 - \delta_\theta) \theta_{1,t-1} \sigma_t / \sigma_{t-1}$$

Non-CO2 forcing law of motion:

$$F_t^{EX} = \min \left(F_{t-1}^{EX} + \Delta_F, F_{CAP} \right)$$

Land-use law of motion:

$$E_t^{\text{land}} = (1 - \delta_e) E_{t-1}^{\text{land}}$$

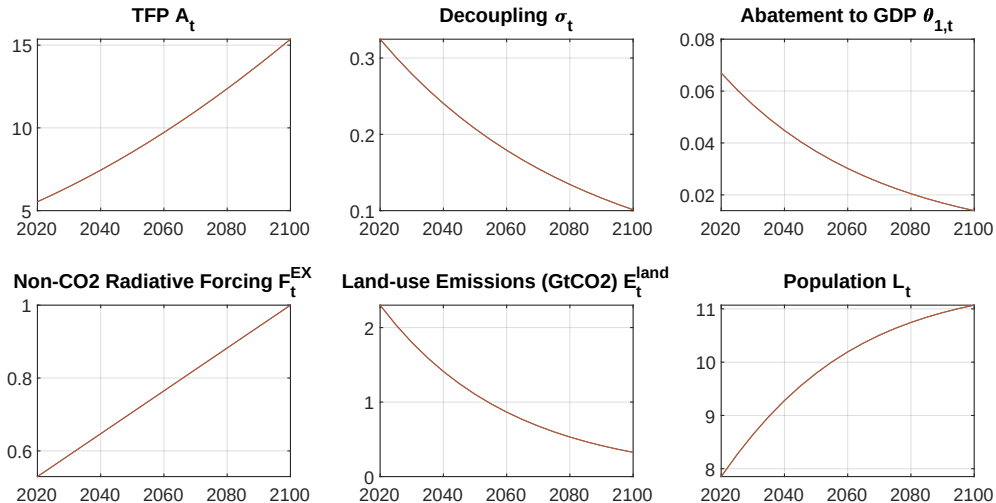


Figure. Exogenous variables

Introducing a saving rate

- ▶ How should we determine optimally C and I ?
- ▶ Could use decentralized decision problem, or use old-fashion centralized equation following Ramsey allocation;
- ▶ Introduce optimal saving rate S , consumption and investment becomes:

$$C_t = (1 - S_t) \times Y_t \left(1 - \theta_{1,t} \mu_t^{\theta_2}\right) \Omega(T_t) \quad (10)$$

$$I_t = S_t \times Y_t \left(1 - \theta_{1,t} \mu_t^{\theta_2}\right) \Omega(T_t) \quad (11)$$

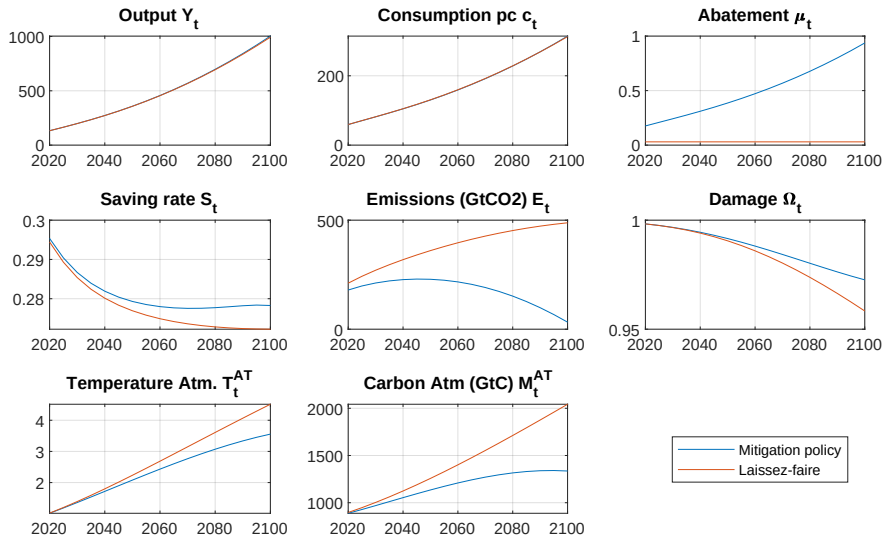
→implicitly assuming that income spent between C_t and I_t .

Introducing a saving rate

- The planner maximizes the social welfare:

$$\max_{\{S_t, \mu_t\}} \sum_{t=0}^{\infty} \beta^t L_t \left(c_t^{1-\alpha} - 1 \right) (1 - \alpha)^{-1} \quad (12)$$

where $c_t = C_t/L_t$ is the per capita consumption, utility function concave.



Some striking but controversial results from DICE:

- ▶ “Optimal” to warm the planet up to 3.5°C ;
- ▶ Mitigation policy saves about 2% of climate damage;
- ▶ Net zero emission optimal by 2100, 50 years later than in Paris-Agreement;
- ▶ Main critic from DICE’s policy output: those results have scientifically grounded climate inaction in policy circles...
... but DICE has also the merit to set the climate issue to the agenda of (macro-)economists.

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→ Let's work on the exercise 1 & 2 of PC2
and baby_rbc1.mlx, baby_rbc2.mlx, baby_rbc3.mlx

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 - ▶ On the one hand, a warming planet causes damages that will make resources scarcer & prices higher → **climateflation**.
 - ▶ On the other hand, the fight against climate change through increasing carbon taxes will increase production costs → **greenflation**.
- ▶ How should the central bank conduct monetary policy in this new landscape?
- ▶ Answering this question requires a new class of IAM with New Keynesian ingredients to capture inflation dynamics.
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This paper

- ▶ The canonical New Keynesian model (e.g. [Woodford2003](#)) not designed for climate analysis.
- ▶ This paper develops The New Keynesian Climate (NKC) model by:
 - ▶ extending with a **carbon accumulation** constraint and a **mitigation policy** from the Integrated Assessment Model (IAM) [Barrage and Nordhaus2023];
 - ▶ estimating NKC for the world economy with techniques that take into account nonlinearities resulting from climate change;
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- ▶ Climate problem: Cumulative emissions permanently alter the propagation mechanism → no steady state.
- ▶ Why? Cumulative carbon induces permanent shifts in productivity, structurally similar to an endogenous (de-)growth model.
- ▶ Climate damages $\Omega(M)$ in production ($Y = \Phi(M) \cdot L$) does not scale proportionally with economic growth, violating the homogeneity assumption required for Solow's detrending techniques.
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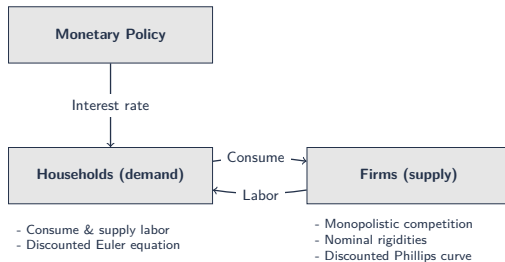
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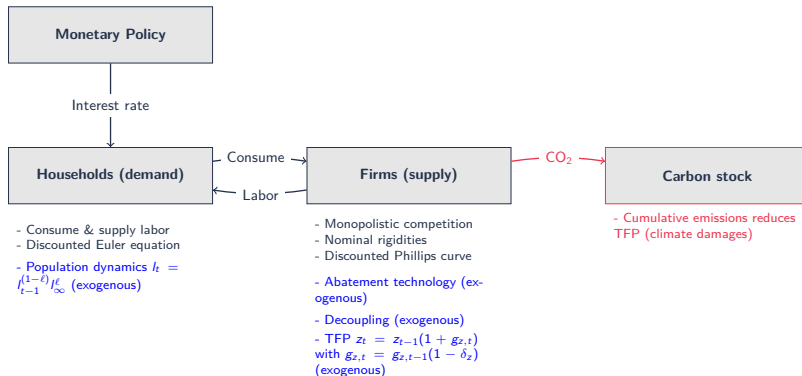
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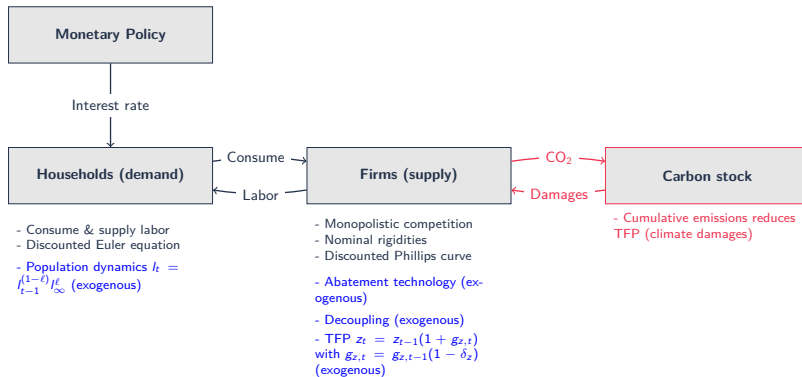
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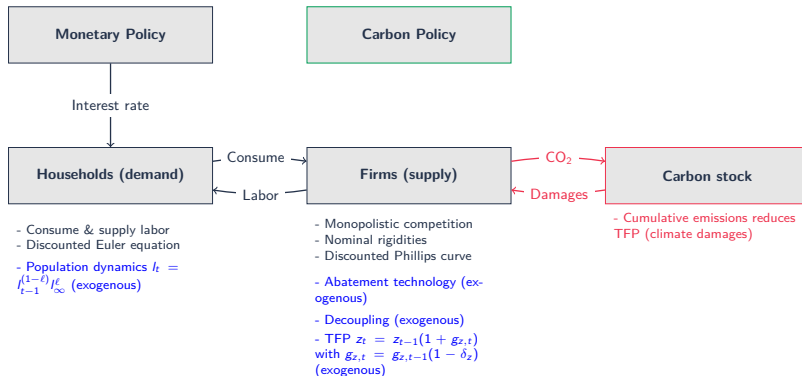
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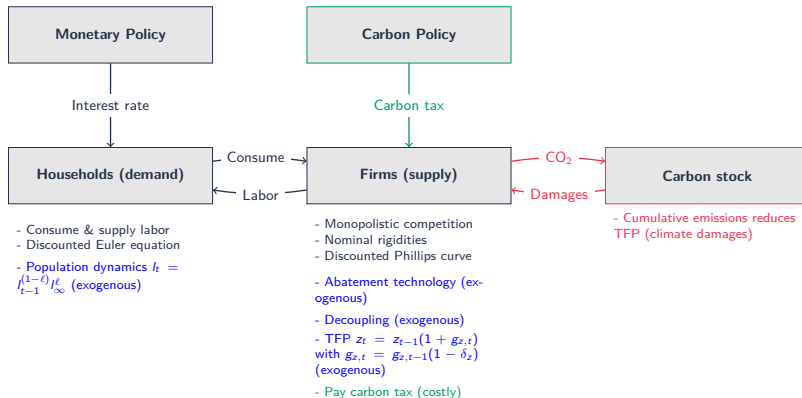
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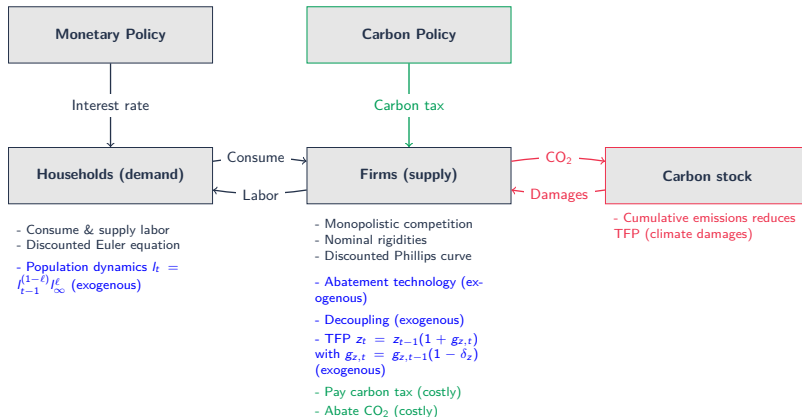
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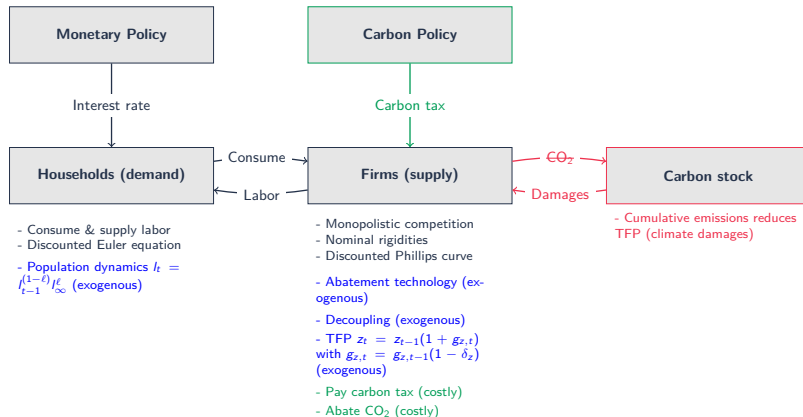
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The New Keynesian Climate Model

Carbon accumulation and its damages:

$$\text{IS:} \quad \left(\frac{\tilde{y}_t x_t - \omega d}{1 - \omega} \right)^{-\sigma_c} = \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1 - \omega) \left(\frac{x_{t+1} \tilde{y}_{t+1} - \omega d}{1 - \omega} \right)^{-\sigma_c} + \omega d^{-\sigma_c} \right)$$

$$x_t = 1 - (1 - \vartheta) 0.5 \kappa (\pi_t - \pi_t^*)^2 - \vartheta (1 - \varepsilon_{p,t} mc_t)$$

$$\text{PC:} \quad (\pi_t - \pi_t^*) \pi_t = (1 - \vartheta) \beta \mathbb{E}_t g_{z,t} \tilde{y}_{t+1} / \tilde{y}_t (\pi_{t+1} - \pi_{t+1}^*) \pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1} (1 - \zeta)$$

$$mc_t = \psi (x_t \tilde{y}_t - \omega d)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha-1} \Phi(\tilde{m}_t)^{-(1+\sigma_n)/\alpha}$$

$$\text{MP:} \quad r_t = r_{t-1}^\rho \left[r_r \pi_t^* (\pi_t / \pi_t^*)^{\phi_\pi} (\tilde{y}_t / \tilde{y}_t^n)^{\phi_y} \right]^{1-\rho} \varepsilon_{r,t}$$

$$\text{CC:} \quad \tilde{m}_t = \tilde{m}_{t-1} + \xi_m \sigma_t z_t l_t \tilde{y}_t \varepsilon_{e,t}$$

The New Keynesian Climate Model

Carbon accumulation and its damages:

$$\text{IS:} \quad \left(\frac{\tilde{y}_t x_t - \omega d}{1 - \omega} \right)^{-\sigma_c} = \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1 - \omega) \left(\frac{x_{t+1} \tilde{y}_{t+1} - \omega d}{1 - \omega} \right)^{-\sigma_c} + \omega d^{-\sigma_c} \right)$$

$$x_t = 1 - (1 - \vartheta) 0.5 \kappa (\pi_t - \pi_t^*)^2 - \vartheta (1 - \varepsilon_{p,t} mc_t)$$

$$\text{PC:} \quad (\pi_t - \pi_t^*) \pi_t = (1 - \vartheta) \beta \mathbb{E}_t g_{z,t} \tilde{y}_{t+1} / \tilde{y}_t (\pi_{t+1} - \pi_{t+1}^*) \pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1} (1 - \zeta)$$

$$mc_t = \psi (x_t \tilde{y}_t - \omega d)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha-1} \Phi(\tilde{m}_t)^{-(1+\sigma_n)/\alpha}$$

$$\text{MP:} \quad r_t = r_{t-1}^\rho \left[r_r \pi_t^* (\pi_t / \pi_t^*)^{\phi_\pi} (\tilde{y}_t / \tilde{y}_t^n)^{\phi_y} \right]^{1-\rho} \varepsilon_{r,t}$$

$$\text{CC:} \quad \tilde{m}_t = \tilde{m}_{t-1} + \xi_m \sigma_t z_t l_t \tilde{y}_t \varepsilon_{e,t}$$



Anthropogenic carbon stock

The New Keynesian Climate Model

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$$\text{MP:} \quad r_t = \left[\frac{\beta \mathbb{E}_t \left[\frac{\varepsilon_{r,t+1}}{\varepsilon_{r,t}} \left(\frac{\tilde{y}_{t+1}}{\tilde{y}_t} \right)^{\phi_y} \right]^{1-\rho}}{\beta \mathbb{E}_t \left[\frac{\varepsilon_{r,t+1}}{\varepsilon_{r,t}} \left(\frac{\tilde{y}_{t+1}}{\tilde{y}_t} \right)^{\phi_y} \right]^{1-\rho}} \right]^{\frac{1}{1-\rho}} \varepsilon_{r,t}$$

$$\text{CC:} \quad \tilde{m}_t = \tilde{m}_{t-1} + \xi_m \sigma_t z_t l_t \tilde{y}_t \varepsilon_{e,t}$$

Deterministic
Decoupling trend

Anthropogenic carbon stock

The New Keynesian Climate Model

Carbon accumulation and its damages:

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Anthropogenic carbon stock

Deterministic

The New Keynesian Climate Model

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$$mc_t = \psi (x_t \tilde{y}_t - \omega d)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha-1} \Phi(\tilde{m}_t)^{-(1+\sigma_n)/\alpha}$$

MP:

r_t

Deterministic
Decoupling trend

Deterministic
population trend

CC:

$\tilde{m}_t$

$$\tilde{m}_t = \tilde{m}_{t-1} + \xi_m \sigma_t z_t l_t \tilde{y}_t \varepsilon_{e,t}$$

Anthropogenic carbon stock

Deterministic
TDP: ...

The New Keynesian Climate Model

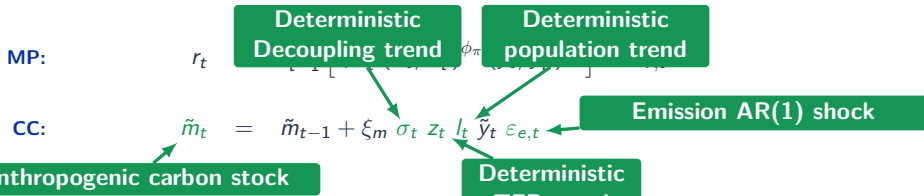
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The New Keynesian Climate Model

Carbon accumulation and its damages:

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$$m c_t = \psi (x_t \tilde{y}_t - \omega d)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha-1} \Phi(\tilde{m}_t)^{-(1+\sigma_n)/\alpha} \leftarrow \text{Climate damages}$$

$$\text{MP: } r_t = r_{t-1}^\rho \left[r_r (\pi_t^* / \pi) (\pi_t / \pi_t^*)^{\phi_\pi} (\tilde{y}_t / \tilde{y}_t^n)^{\phi_y} \right]^{1-\rho} \varepsilon_{r,t}$$

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The New Keynesian Climate Model

Mitigation policies as function of exogenous carbon tax $\tilde{\tau}_t$:

$$\text{IS:} \quad \left(\frac{\tilde{y}_t x_t - \omega d}{1 - \omega} \right)^{-\sigma_c} = \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1 - \omega) \left(\frac{x_{t+1} \tilde{y}_{t+1} - \omega d}{1 - \omega} \right)^{-\sigma_c} + \omega d^{-\sigma_c} \right)$$

Mitigation
expenditures

$$x_t = 1 - (1 - \vartheta) 0.5 \kappa (\pi_t - \pi_t^*)^2 - \vartheta (1 - \varepsilon_{p,t} mc_t) - \theta_{1,t} \tilde{\tau}_t^{\theta_2 / (\theta_2 - 1)}$$

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$$mc_t = \psi (x_t \tilde{y}_t - \omega d)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha-1} \Phi(\tilde{m}_t)^{-(1+\sigma_n)/\alpha} + \theta_{1,t} \tilde{\tau}_t (\theta_2 + (1 - \theta_2) \tilde{\tau}_t^{1/(\theta_2-1)})$$

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The New Keynesian Climate Model

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Carbon tax costs

$$\text{PC:} \quad (\pi_t - \pi_t^*) \pi_t = (1 - \vartheta) \beta \mathbb{E}_t g_{z,t} \tilde{y}_{t+1} / \tilde{y}_t (\pi_{t+1} - \pi_{t+1}^*) \pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1} (1 - \zeta)$$

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The New Keynesian Climate Model

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Mitigation
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$$x_t = 1 - (1 - \vartheta) 0.5 \kappa (\pi_t - \pi_t^*)^2 - \vartheta (1 - \varepsilon_{p,t} mc_t) - \theta_{1,t} \tilde{\tau}_t^{\theta_2 / (\theta_2 - 1)}$$

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Abatement
share

$$\text{CC:} \quad \tilde{m}_t = \tilde{m}_{t-1} + \xi_m \sigma_t z_t l_t \tilde{y}_t \varepsilon_{e,t} \left(1 - \tilde{\tau}_t^{1 / (\theta_2 - 1)} \right)$$

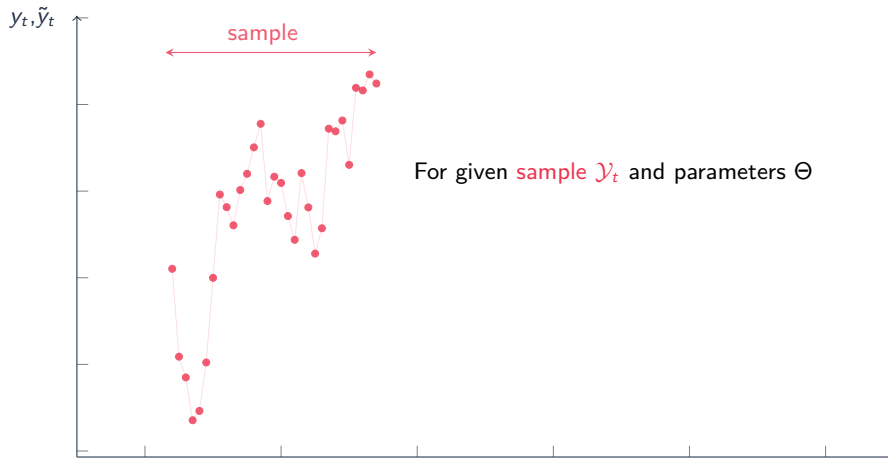
Estimation

- ▶ Estimation on world data from 1985Q1 to 2023Q3 (sources: World Bank, OECD and OurWorldInData).
- ▶ There are four observable variables:

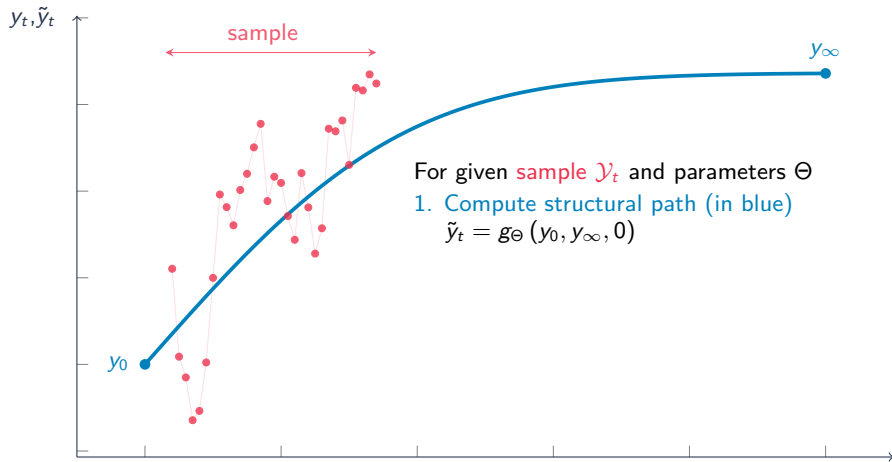
$$\begin{bmatrix} \text{Real output growth rate} \\ \text{Inflation rate} \\ \text{Short-term interest rate} \\ \text{CO}_2 \text{ emissions growth rate} \end{bmatrix} = 100 \times \begin{bmatrix} \Delta \log(y_t) \\ \pi_t - 1 \\ r_t - 1 \\ \Delta \log(e_t) \end{bmatrix}$$

- ▶ Solution & filtering methods from [Fair and Taylor1983](#)): fully nonlinear, MIT shocks and no aggregate uncertainty.

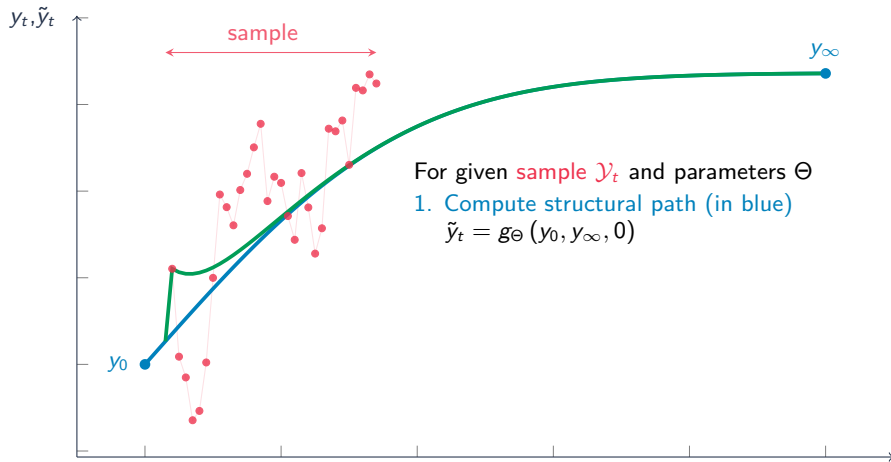
Estimation



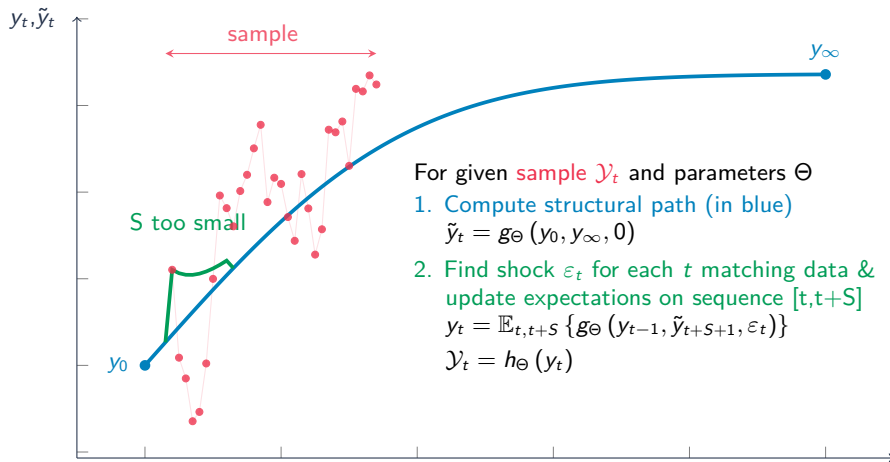
Estimation



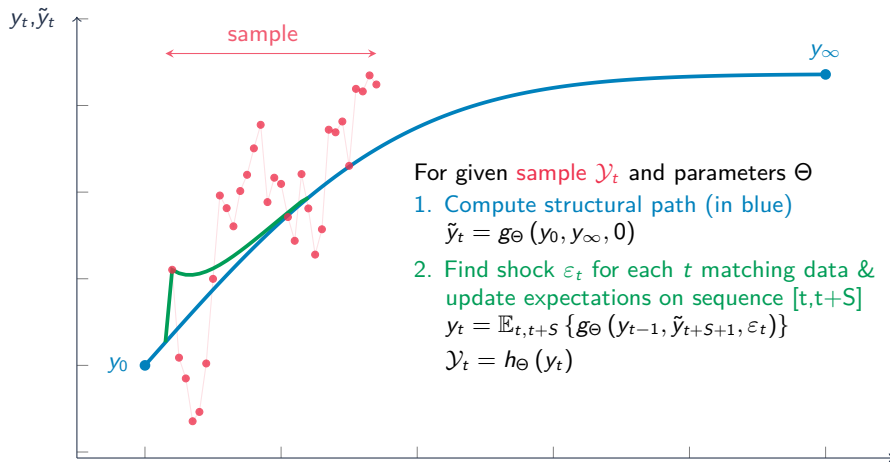
Estimation



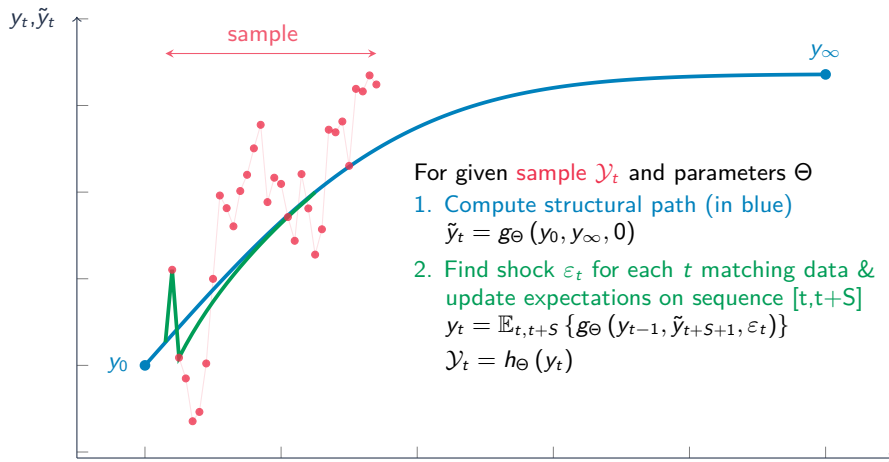
Estimation



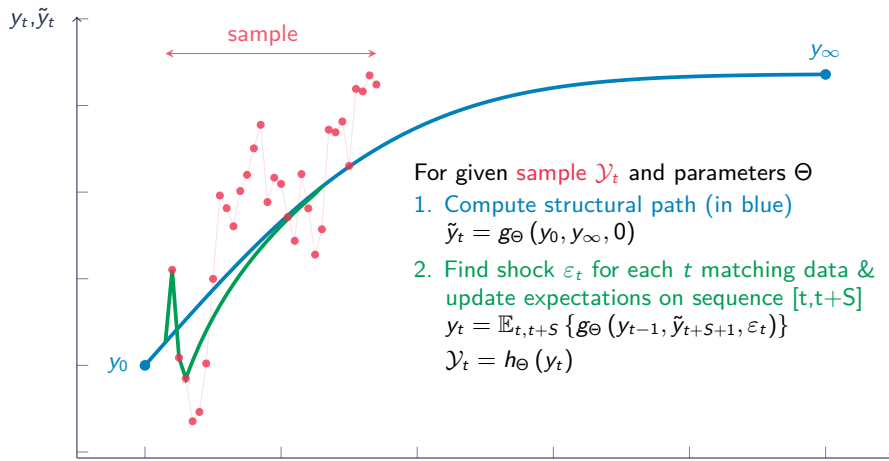
Estimation



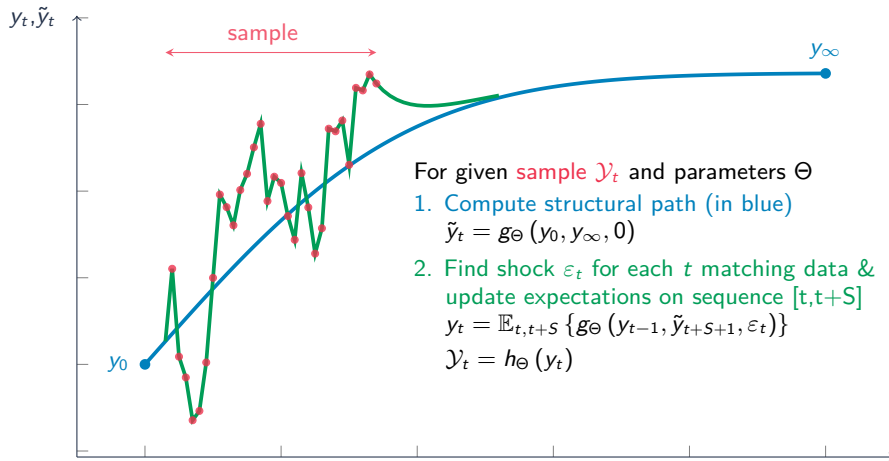
Estimation



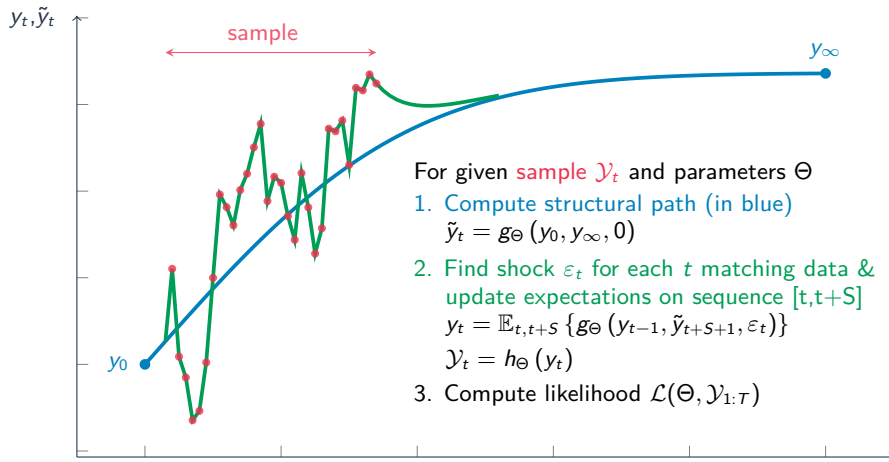
Estimation



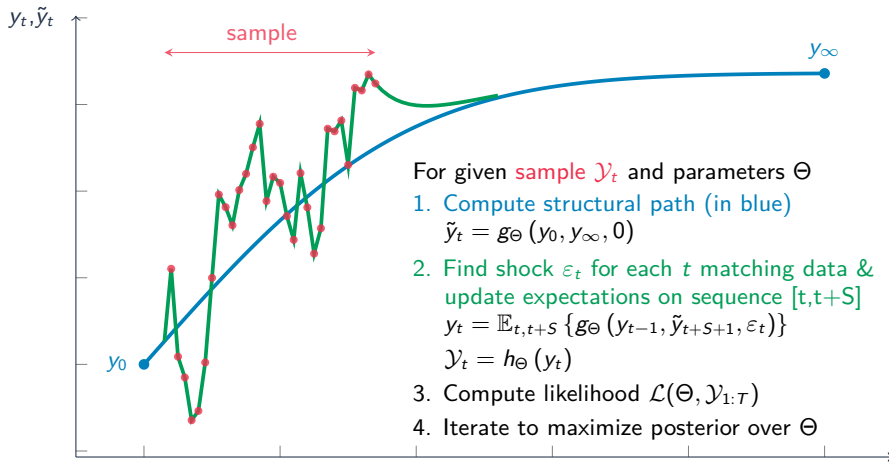
Estimation



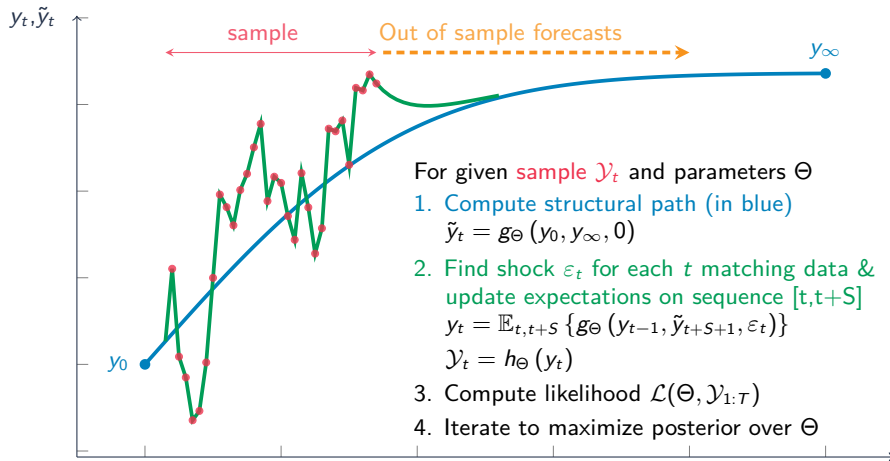
Estimation



Estimation



Estimation

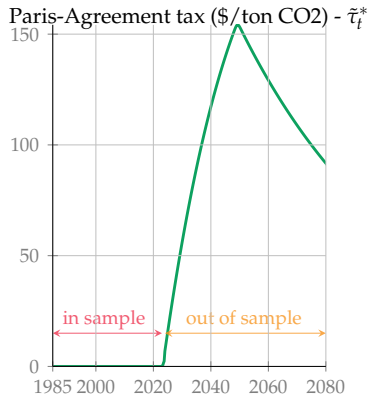


Estimation

- Large uncertainty about future carbon tax: implications for estimation in particular at the end of the sample
- Let $\tilde{\tau}_t^*$ denote the Paris-Agreement tax, with linear decrease in carbon emissions up to 2050
- We let the data inform about future carbon mitigation policies:

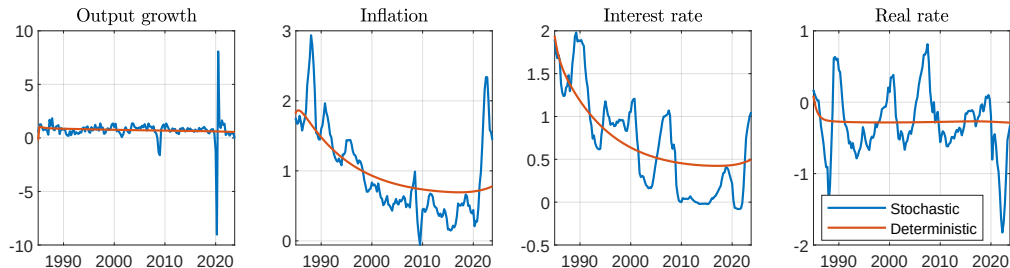
$$\mathbb{E}_t\{\tilde{\tau}_t\} = \varphi \tilde{\tau}_t^*$$

where $\varphi \in [0, 1]$ is the fraction of believers (or expected intensity) of Paris-Agreement policy.



Stochastic and deterministic paths

Figure 1: Implied deterministic and stochastic paths

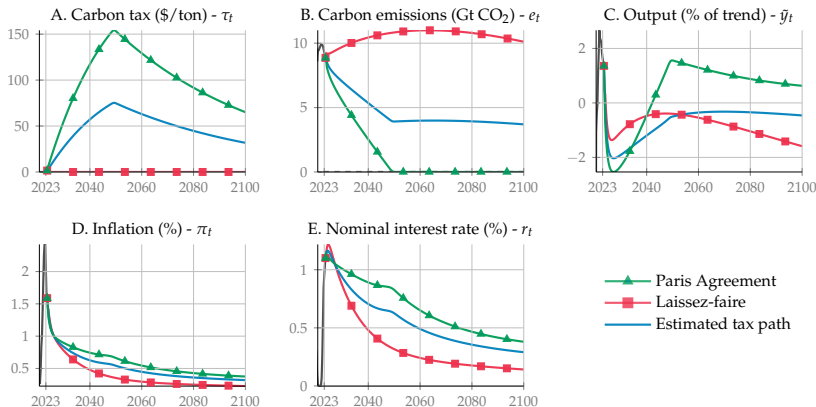


The Anatomy of Green/Climateflation

- ▶ What is the future macroeconomic landscape by the end of the century?
- ▶ We consider three alternative scenarios based on the realization of the carbon tax $\varphi \tilde{\tau}_t^*$:
 - ▶ Paris-Agreement with $\varphi = 1$
 - ▶ Estimated carbon path with $\varphi = 0.53$
 - ▶ Laissez-faire with $\varphi = 0$

Three Transitions

Figure 2: Model-implied projections based on alternative control rates of emissions



Dissecting the PC curve

- Stabilization objective of a central bank: important to understand how climate affects inflation.
- One can split the marginal cost into three term:

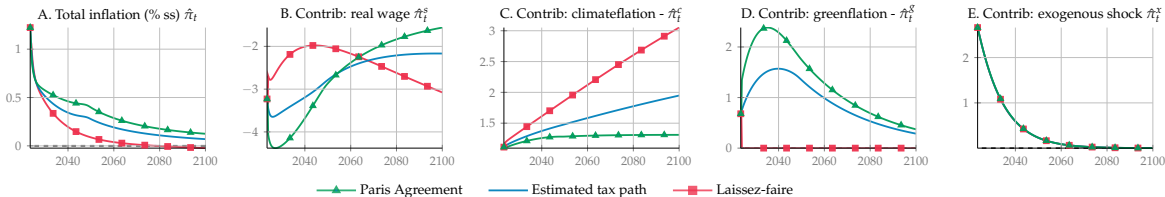
$$mc_t = \underbrace{\tilde{w}_t}_{\text{real wage}} / \underbrace{\Phi(m_t)}_{\text{climateflation}} + \underbrace{\theta_{1,t}\mu_t^{\theta_2} + \tau_{e,t}\sigma_t(1-\mu_t)\varepsilon_{e,t}}_{\text{greenflation}}, \quad (13)$$

which allows to break down inflation into 4 different forces:

$$\hat{\pi}_t \simeq \underbrace{\hat{\pi}_t^S}_{\text{wage term}} + \underbrace{\hat{\pi}_t^C}_{\text{climateflation}} + \underbrace{\hat{\pi}_t^G}_{\text{greenflation}} + \underbrace{\hat{\pi}_t^X}_{\text{exogenous shocks}} \quad (14)$$

with $\hat{\pi}_t = \pi_t - \pi_t^*$

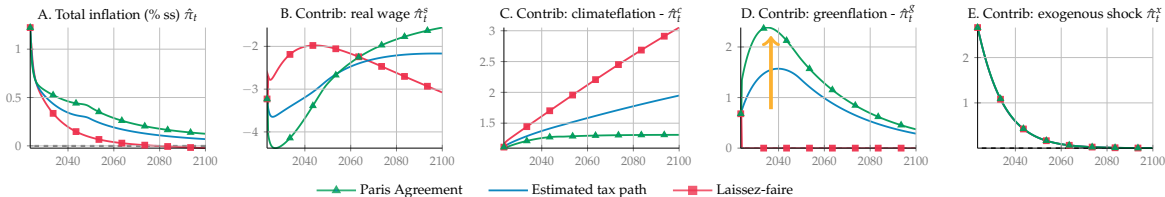
Dissecting the PC curve



► Under Paris Agreement:

- The immediate increase in carbon tax fuels inflation
- General equilibrium effect: increasing abatement expenditures reduces both consumption and in turn the wealth effect on the labor supply
- Net zero stabilizes damages, and hence climateflation

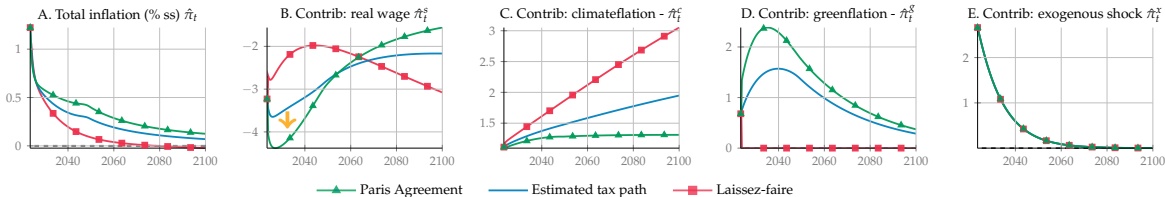
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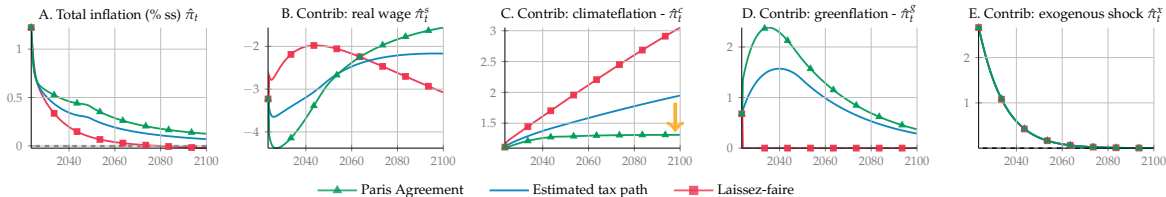
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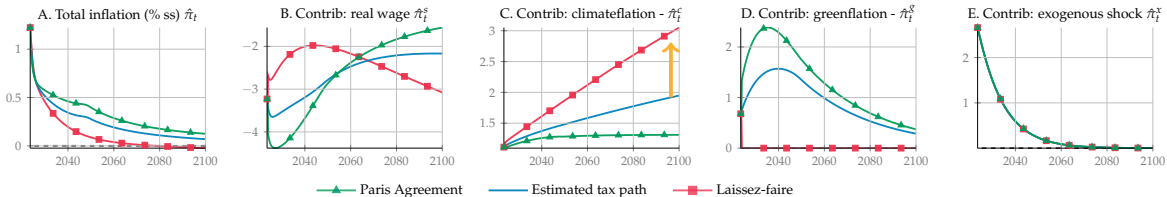
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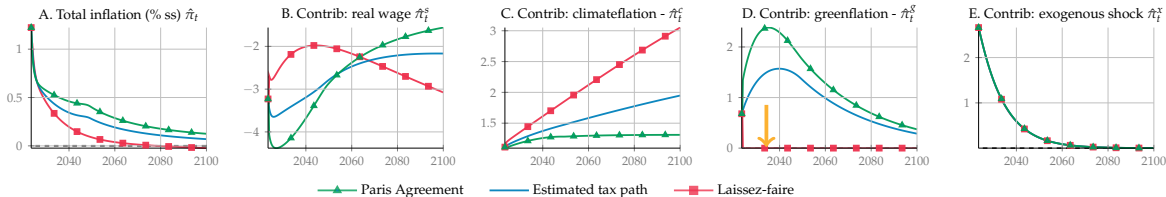
Dissecting the PC curve



► Under Laissez-faire:

- The rising damage makes resources scarcer: ever growing inflation as long as planet warms
- Disengagement from carbon policy makes carbon price to be zero
- General equilibrium effect: real wages fall as climate decreases productivity

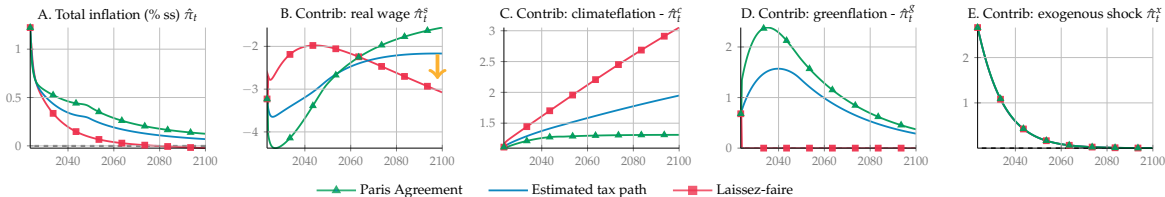
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→ Let's work on `/Codes2/C_NK/Notebook_NK.mlx`

Thank you for your attention

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